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(57) Abstract :
Urban flooding makes real-time monitoring difficult, causing delayed response and damage. Traditional methods are slow, costly, and often inaccurate. This project proposes an AI-based flood detection system using CCTV cameras and YOLOv8 in Python to detect water levels from live video feeds. A Raspberry Pi 4 is used to host the web dashboard continuously in 24/7 mode for real-time monitoring. The system stores flood detection details in a XAMPP MySQL database for analysis and record maintenance. This cost-effective solution uses existing infrastructure, live alerts, and database support to improve disaster management.

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